

Quiz 2: Solutions

Give exact answers and justify your reasoning. Do not use electronic tools. Use the limit definition, uniqueness, and the sum law covered in Lecture 3; do not assume other limit laws or continuity theorems. All functions are real-valued. Unless otherwise specified, a function whose limit at a is considered is defined on an open interval containing a , except possibly at a itself. Every condition on x refers to points in the domain of the function or the common domain of the functions involved.

1. Write the epsilon-delta definition of $\lim_{x \rightarrow -1} f(x) = 2$, and then write its negation.

Solution.

Let D be the domain of f . The definition is: for every $\varepsilon > 0$, there exists $\delta > 0$ such that, for every $x \in D$,

$$0 < |x + 1| < \delta \implies |f(x) - 2| < \varepsilon.$$

Its negation is: there exists $\varepsilon_0 > 0$ such that, for every $\delta > 0$, there exists $x \in D$ with

$$0 < |x + 1| < \delta \quad \text{and} \quad |f(x) - 2| \geq \varepsilon_0.$$

2. Prove directly from the definition that $\lim_{x \rightarrow -1} x^2 = 1$. First find a constant bound for $|x - 1|$ under the restriction $|x + 1| < 1$.

Solution.

Under the restriction $|x + 1| < 1$, the triangle inequality gives

$$|x - 1| = |(x + 1) - 2| \leq |x + 1| + 2 < 3.$$

Let $\varepsilon > 0$ and choose

$$\delta = \min\{1, \varepsilon/3\} > 0.$$

If $0 < |x + 1| < \delta$, then $|x + 1| < 1$, so the bound above applies. Consequently,

$$|x^2 - 1| = |x + 1| |x - 1| < 3|x + 1| < 3\delta \leq \varepsilon.$$

This proves $\lim_{x \rightarrow -1} x^2 = 1$ directly from the definition.